

# Embedding Spaces handout — corrections and measured figures

Checked against the Final Project Report (68 pp) and G10's executed output. Every figure in Part B is a measured number quoted from a named table; nothing here is simulated, interpolated or re-derived.

## Part A — corrections

The handout is accurate on the conceptual material: the evidence ladder, the geometry/topology distinction, the kNN and CKA definitions, the ridge/CCA/Procrustes progression, the retrieval-versus-verification failure modes, and the  $N = 9,533 = 8,533 + 1,000$  split. The corrections below are all about hub composition and about which diagnostics have already been run.

| where                                    | the handout says                                                                                                                                      | correction                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pp. 7, 11, 12, 13<br>(the central claim) | the documented G1 Hub is FIVE arms (3 DINO + BGE + GPT-2, 6144-d); BERT and SBERT are not Hub inputs; it "should not be described as a seven-arm Hub" | THREE configurations exist and are being collapsed into one. The 6144-d five-space hub is what Section C.13's BODY describes. The G1 notebook builds SEVEN spaces (7680-d). The published 94.2 percent belongs to a FOUR-space hub (3 DINO + bge, 5376-d), reproduced exactly by G0-exact. G10's own variant labels confirm it: 'bge [= G4 hub]' is four spaces, 'all four [= G1 hub]' is seven. |
| p. 12<br>"Option 1"                      | adding BERT and SBERT to the frozen Hub is presented as a proposed extension                                                                          | already done, and documented. Report p. 43 (C.13.3): adding BERT and Sentence-BERT "brings it to seven spaces spanning FOUR text training objectives". Note it was a REBUILD, not a frozen-hub add: the PCA basis was recomputed over seven spaces.                                                                                                                                              |
| p. 11<br>scope table                     | BERT and SBERT scoped as "7-space tests" only                                                                                                         | both are documented hub members after C.13.3. SBERT additionally serves as a HEAD TARGET in G9 and in G10's Q2 controls, so it occupies all three roles: hub member, source, and target.                                                                                                                                                                                                         |
| p. 4<br>"Project status"                 | multi-k neighbourhood overlap "should be computed" across the 21 pairs                                                                                | already computed. Table C12h gives $k = 1/10/100/500$ as a ratio to chance for four reference pairs; CKNN is computed across $k$ for all 21 pairs, 21 of 21 scoring higher locally than at the global limit (C.13.5, C.13.9). See Figure 1 overleaf.                                                                                                                                             |
| p. 10<br>"proposed diagnostics"          | multi-scale kNN overlap, distance-rank agreement and CCA listed as complementary or proposed                                                          | all three have been run. CCA shared-direction counts are Table C12k (cross-modal pairs share 6–32 of 64 directions; image siblings and same-objective text pairs share nearly all). Distance-rank agreement is Spearman +0.681 / Pearson +0.763 (Table C10b).                                                                                                                                    |
| throughout                               | no mention of hub composition sensitivity or of the target-in-hub dependency                                                                          | both now measured and declared. G10: text-member IDENTITY moves transfer 6.1 pt at fixed count, presence-vs-absence moves it 11.2 pt; the head's target sitting inside its own hub is worth +3.0 pt beyond the generic benefit of any text member. Declared in the report at C.13.2 and E.6. See Figures 2 and 3.                                                                                |

## The three hub configurations, stated once

| configuration | spaces | concat width | what it is                                                          |
|---------------|--------|--------------|---------------------------------------------------------------------|
| G4 / G0-exact | 4      | 5,376        | 3 DINOv2 + bge-m3. The published 94.2 percent cross-lineage figure. |
| C.13 body     | 5      | 6,144        | adds GPT-2. What the handout's 6144-d diagram actually depicts.     |
| C.13.3 / G1   | 7      | 7,680        | adds BERT and SBERT. Four text objectives, two modalities.          |

## On UMAP

Not recommended as evidence, and the reason is structural rather than aesthetic. UMAP's objective function IS local-neighbourhood preservation, so using it to demonstrate that encoders agree locally is circular: the method manufactures the appearance of the finding, and does so even on data with no shared structure. It distorts global geometry by construction — exactly the axis this project makes claims about — and it is stochastic, so "what happens under a different seed?" is a fair and awkward question. The local-over-global conclusion already rests on four independent instruments (kNN ratio-to-chance, CKNN, distance-rank correlation, raw cross-space cosine at zero); adding a UMAP panel would widen the attack surface without adding evidence. Acceptable as a clearly-labelled illustration in a talk; not in a technical document.

## Part B — measured figures

Each figure carries the table or experiment it is drawn from. These are the plots the handout is missing; none require access to the vector caches.

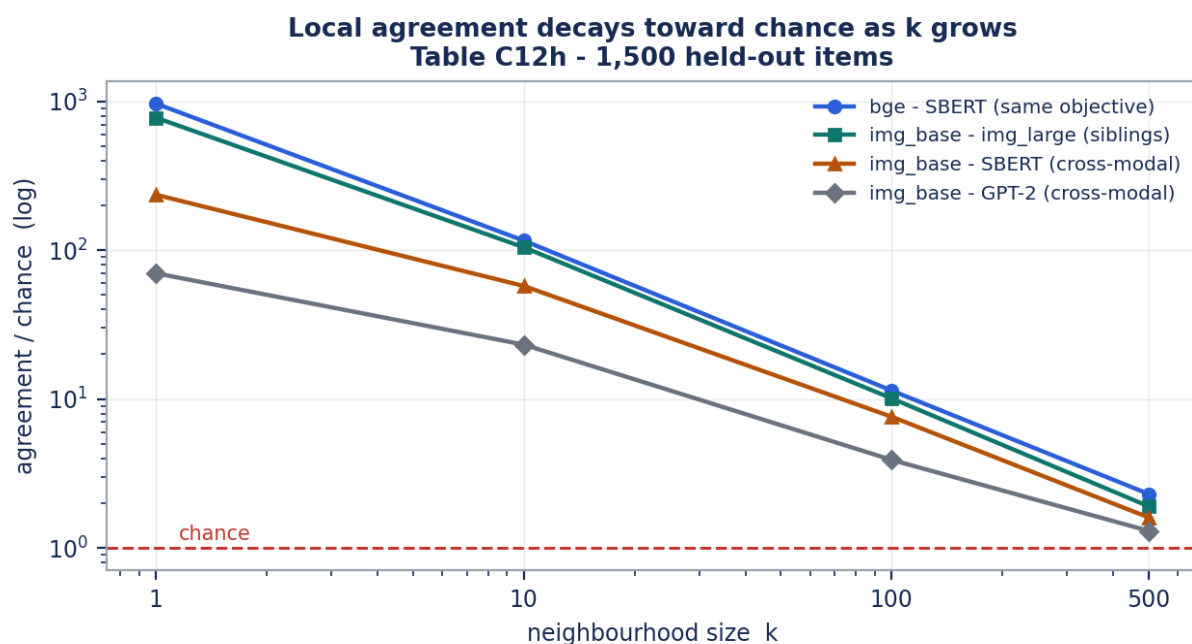


Figure 1. The local-over-global finding, plotted. Every pair starts far above chance at  $k = 1$  and converges toward it by  $k = 500$ , and the ordering — same-objective text, image siblings, cross-modal, GPT-2 — is stable at every scale. Source: Table C12h, 1,500 held-out items.

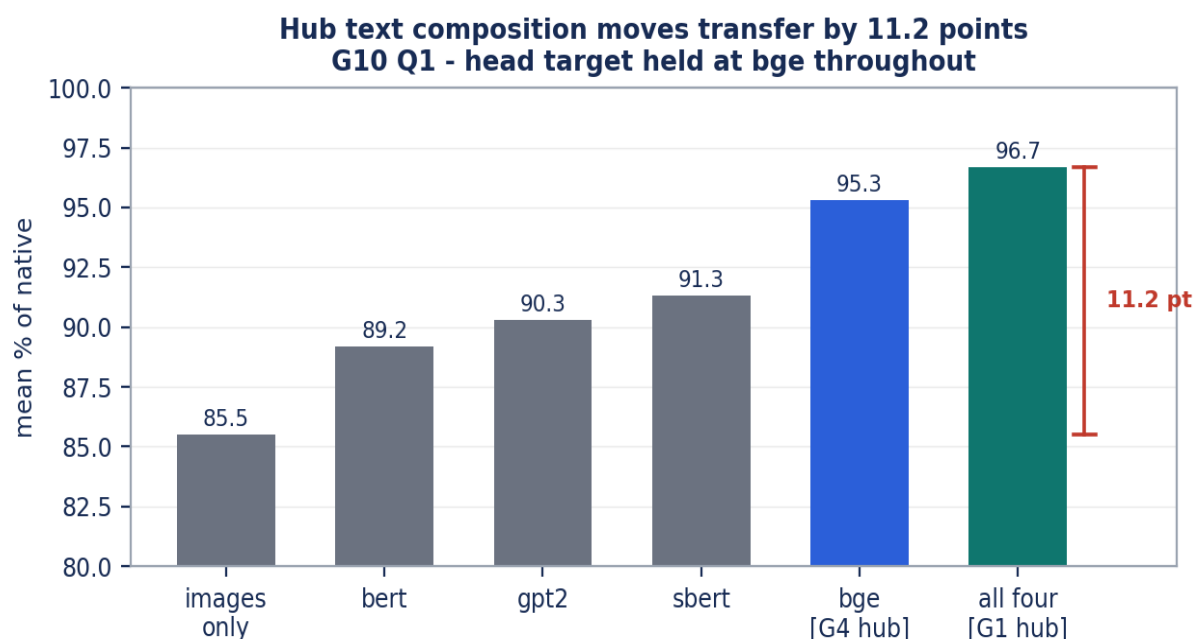


Figure 2. Hub composition is not a free choice. Holding the head's target fixed at bge, swapping which text space sits in the hub moves transfer 6.1 pt at fixed count (bge 95.3 vs bert 89.2), and presence versus absence spans 11.2 pt. This is what qualifies E.6's low-reach reading. Source: G10 Q1.

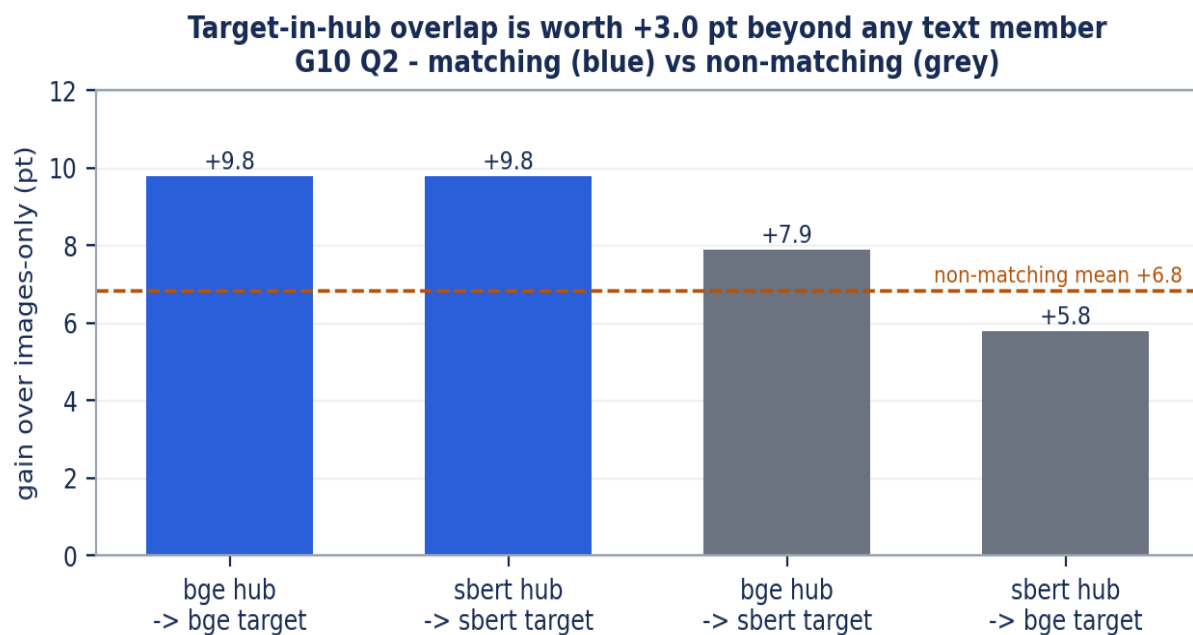


Figure 3. The target-in-hub dependency, isolated. Matching cases (blue) gain +9.8 pt over images-only; non-matching cases (grey) gain +6.8 pt on average. The difference, +3.0 pt, is what having the head's own target inside the hub is worth. Held-out scoring throughout, so this is a dependency, not leakage. Source: G10 Q2.

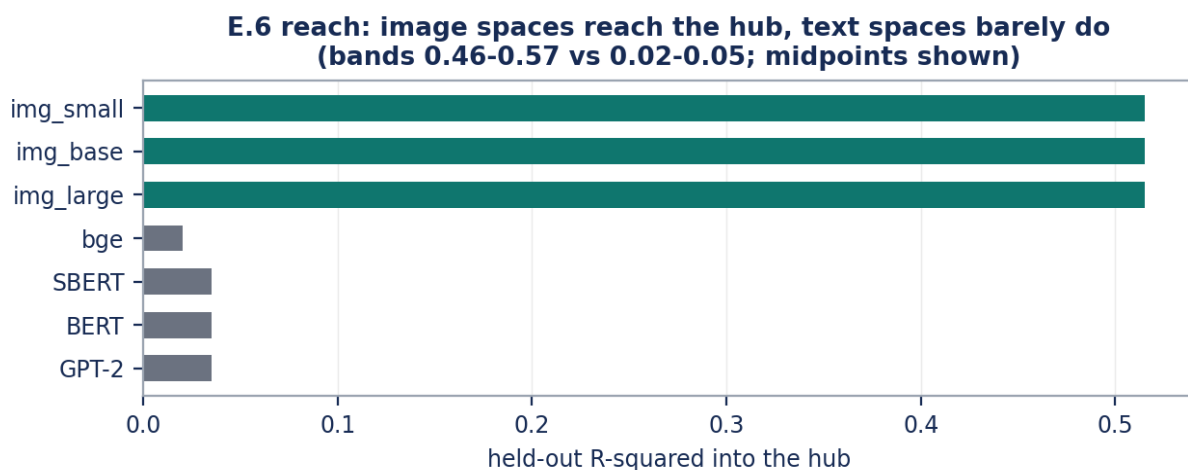


Figure 4. Why the hub is called an image-side object. Image encoders reach R-squared 0.46–0.57 of the hub from their own space; all four text spaces reach 0.02–0.05, uniformly, with collapsed GPT-2 indistinguishable from healthy SBERT. Figure 2 is the necessary caveat to this one. Source: E.6.

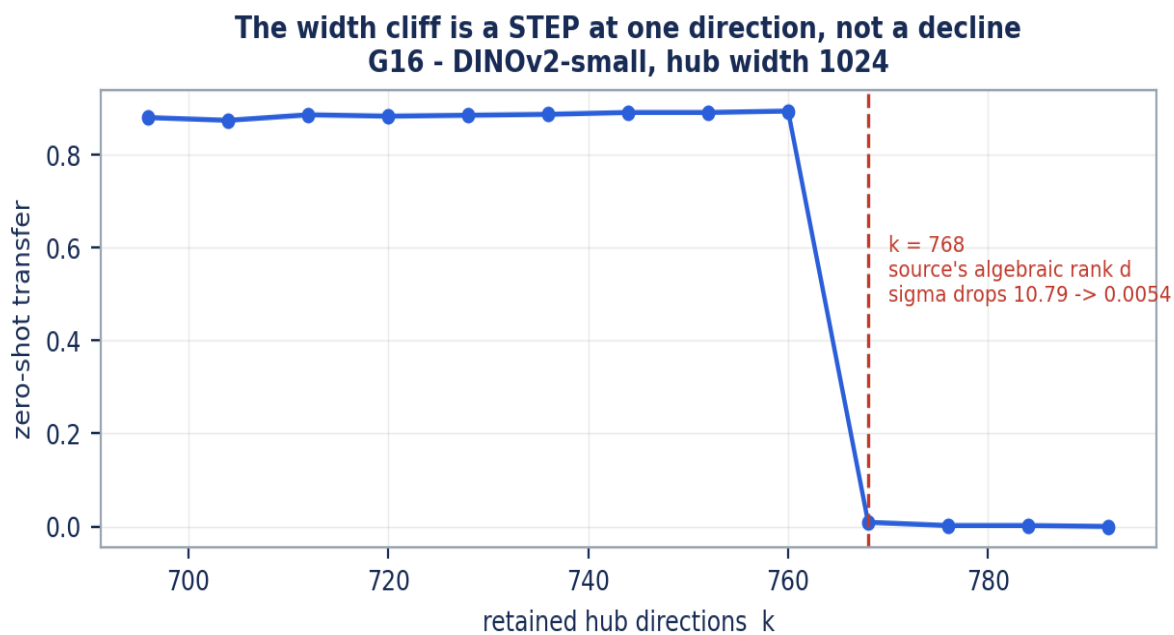


Figure 5. The width cliff resolved to a single direction. Transfer holds above 0.874 through  $k = 760$ , then falls to 0.009 at  $k = 768$  — the source encoder's algebraic rank, where the last retained singular value drops from 10.79 to 0.0054 and least squares divides by it. A step, not a decline. Source: G16.

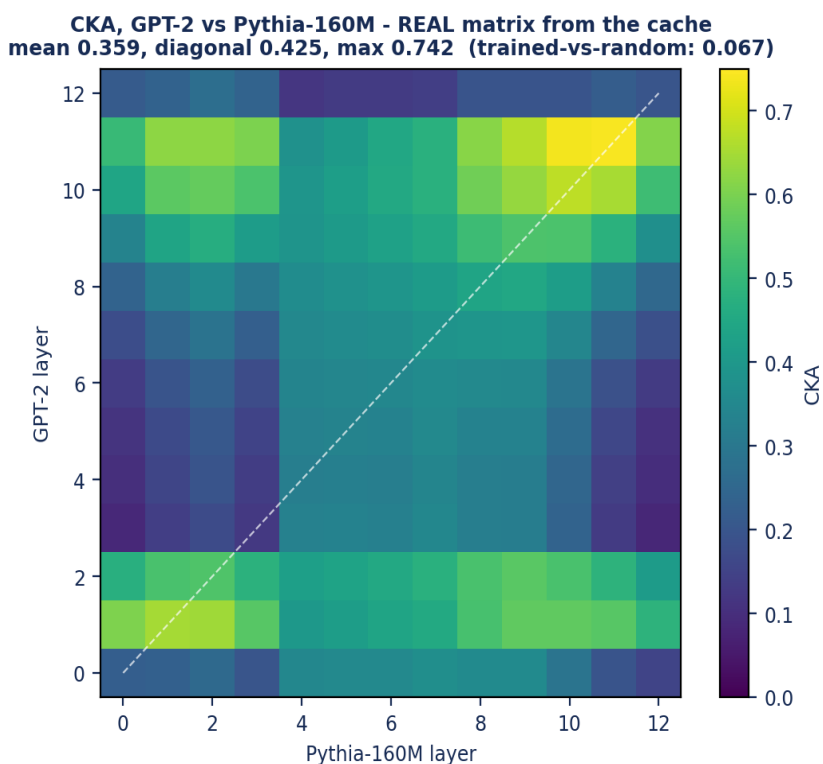


Figure 6. The raw CKA matrix itself, GPT-2 vs Pythia-160M — rendered directly from the cached `cka_matrices.npz` rather than from quoted numbers (this cached run: mean 0.359, diagonal 0.425, max 0.742; the report quotes 0.424 / 0.744 from its final run, noting 'essentially unchanged from run 1'; trained-vs-random 0.067, a 4–5x learning effect). Early layers match early, late match late, and the final layer stands apart — the characteristic pattern the report describes in prose, now visible. Source: Experiment A2 cache.

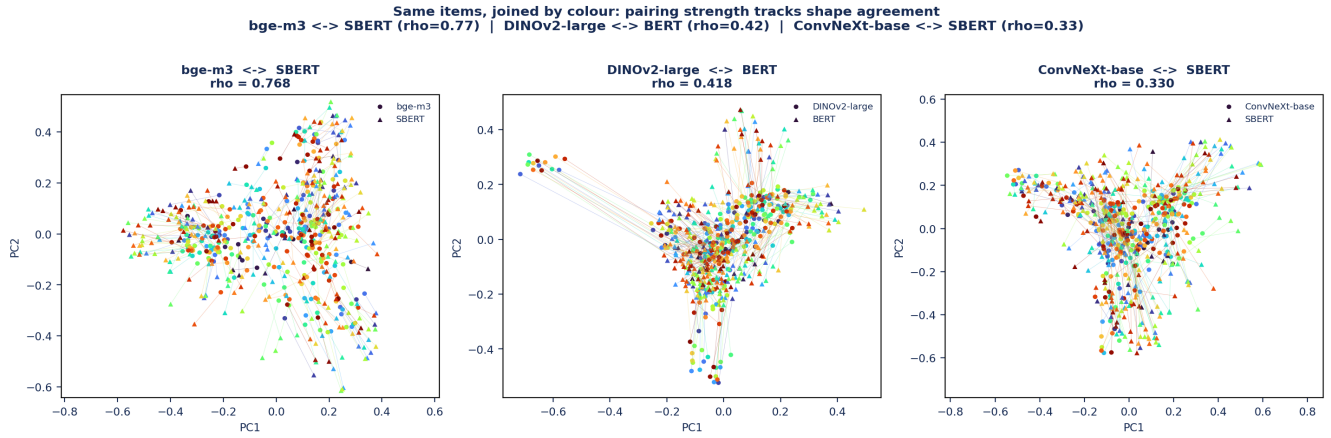


Figure 7a. Same items across three encoder pairs, joined by colour — computed from the raw cached vectors (hub\_rebuilt.npz, 300-item sample, seeded). Pairs auto-selected as the strongest in each class. Left: bge-m3 vs SBERT ( $\rho = 0.768$ , text vs text) — tie-lines short, items land together. Middle: DINOv2-large vs BERT ( $\rho = 0.418$ , strongest DINOv2-text pair). Right: ConvNeXt-base vs SBERT ( $\rho = 0.330$ , strongest ConvNeXt-text pair) — loosest of the three. Pairing tightness tracks  $\rho$  monotonically. Every dot is a real COCO item; each encoder PCA'd to 2-D separately, then Procrustes-aligned, since the encoders differ in width.

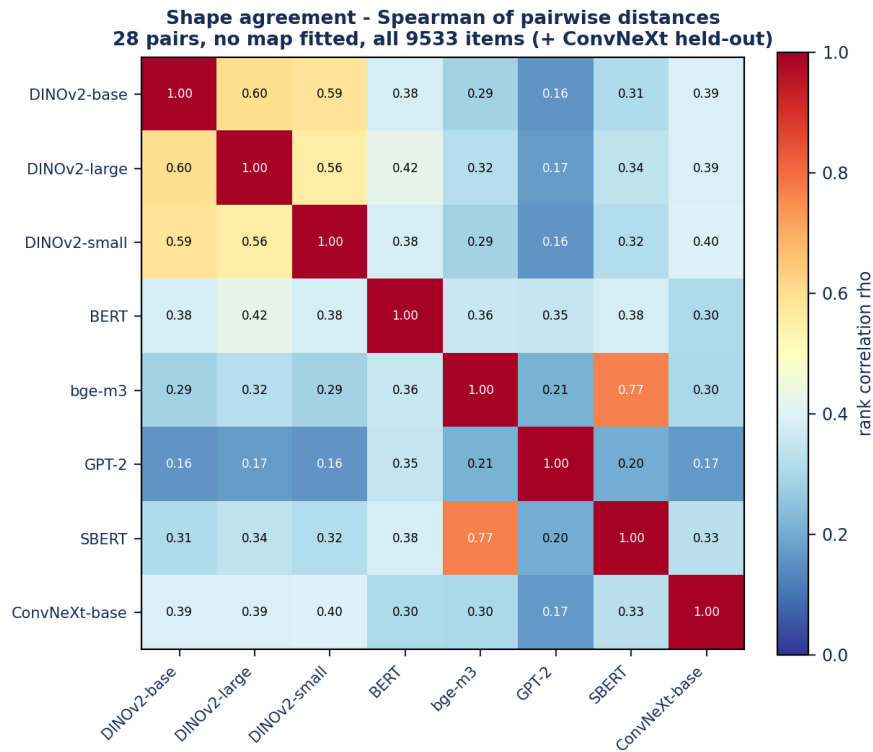


Figure 7b. The shape-agreement matrix, computed live on all 9,533 items from the raw cached vectors, now extended to EIGHT encoders (28 pairs) by adding ConvNeXt-base, which was held out of the hub. No map is fitted anywhere. The seven hub members reproduce Fig A15b: bge-m3–SBERT tops at 0.768, cross-modal mean 0.290, image siblings 0.56–0.60, collapsed GPT-2 the coldest row. ConvNeXt's new row is the finding: 0.30–0.40 against everything, the LOWEST of any image encoder — yet it is the best-transferring encoder through the hub at 96.7 percent. Raw agreement does not predict transfer (report E.12). Source: hub\_rebuilt.npz + ConvNeXt cache, Spearman over 45.4M pairwise distances each.

## Part C · PCA and whitening (see the standalone visual guide)

**The full step-by-step is a separate 10-page deck:** PCA\_Cat\_Dog\_All\_Graph\_States.pdf · the same cat/dog points carried through original -> centred -> eigenvectors -> projection -> rotation -> eigenvalues -> whitening, one transformation per page. This note records the definitions and the answers to the three questions that guide asks.

**PCA.** Directions of maximum variance. EIGENVECTOR = direction; EIGENVALUE = variance along it; the directions are orthogonal and sorted. In the deck's example PC1 eigenvalue 4.52, PC2 0.17 · PC1 carries 96.4% of the variance. Projecting onto them is a pure rotation: shape unchanged, axes realigned.

**Whitening.** Divide each PCA coordinate by the square root of its eigenvalue, so every direction ends at variance 1 and the cloud becomes round. It rescales the axes, not the points: cats and dogs stay separate · only the frame is equalised. This is why whitening rescues cosine retrieval (C.11) yet destroys shape-matching (C.12): the uneven variance it removes is noise for reading a known correspondence and signal for finding an unknown one.

**Q1 · Does PCA separate cats and dogs?** Not by itself · PCA is unsupervised, finding variance not class. Here the clusters are offset, so most separation lands on PC1; that is a property of the data, not a guarantee of PCA.

**Q2 · How does whitening affect the features?** It equalises their scales, so a distinction in a low-variance direction is no longer drowned by a high-variance one · the C.11 rescue mechanism (GPT-2 R@1 0.130 -> 0.478).

**Q3 · Next steps for classification?** A classifier on whitened coordinates weighs all directions evenly; whether that helps depends on where the class signal sat. The project's sharper lesson: the SAME whitening fixes reading (retrieval) and harms finding (unsupervised matching) · opposite prescriptions, one geometry, both measured.

*Correction carried into the deck (State 7): whitening does NOT make cats and dogs identical · it makes the cloud round, changing coordinates, never which items are distinct.*